

Md. Abdur Rahman

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Education

Military Institute of Science and Technology (MIST) — *M.Sc. Engg in Computer Science and Engineering* Apr 2026 – Present

Southeast University — *BSc in Computer Science and Engineering* Oct 2021 – Oct 2025
CGPA: 3.87/4.00

Relevant Coursework: Artificial Intelligence, Image Processing, Data Mining, Research Methodology

Experience

Research Assistant — *Islamic University of Technology (IUT), OIC* Gazipur, BD
Nov 2025 – Present

- Engineered a Federated Learning framework for fintech data with a novel aggregation technique for non-IID settings.
- Pioneered hyperbolic Graph Neural Networks for blockchain fraud detection and causal inference methods for high-dimensional auditing.
- Authored five manuscripts accepted in Q1 journals on decentralized AI, graph-based fraud detection.

Research Assistant — *CAIR, Southeast University* Dhaka, BD
Jan 2025 – Oct 2025

- Designed hybrid transformer-based architectures for medical image segmentation and classification, fusing convolutional and attention mechanisms to improve diagnostic accuracy on MRI and X-ray datasets.
- Integrated explainable AI methods (SHAP, LIME) into clinical decision-support pipelines, enabling interpretable model outputs for high-stakes diagnostic settings.
- Authored five papers: three accepted at peer-reviewed conferences, two under review in Q1 journals on medical AI and interpretable deep learning.

Selected Publications

J = Journal *C* = Conference *W* = Workshop

[J.1] **MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification** [Paper]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Farzan Majeed Noori, Md Zia Uddin, Abdul Kadar Muhammad Masum
Frontiers in Plant Science [Q1] [🔗](#)

[J.2] **Integrating Domain-Specific Convolutional Features with Swin Transformers for Transparent Cervical Cancer Diagnostics** [Paper]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Md Fokrul Islam KHAN, Md Zia Uddin, Ulysse Teller Masao Côté-Allard, Abdul Kadar Muhammad Masum
Frontiers in Oncology [Q2] [🔗](#)

[J.3] **FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection** [Paper]

Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Md Fokrul Islam Khan, Farzan M. Noori, Abdul Kadar Muhammad Masum
Frontiers in Public Health [Q1] [🔗](#)

[J.4] **DFU-GCNet: A Global Context-Enhanced Inception Network for Robust and Interpretable Diabetic Foot Ulcer Classification** [Paper]

Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Farzan M. Noori, Abdul Kadar Muhammad Masum
Frontiers in Digital Health [Q1] [🔗](#)

[J.5] **Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles** [Paper]

Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, **Md. Abdur Rahman**, Md. Tofael Ahmed Bhuiyan, Ramona Birău, Virgil Popescu
IEEE Access [Q1] [🔗](#)

[J.6] **Predicting Corporate Bankruptcy Through a Scalable and Interpretable Knowledge Distillation Framework** [Paper]

Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, **Md. Abdur Rahman**, Md. Tofael Ahmed Bhuiyan, Thanh Lam Nguyen, Hasan Dinger, Serhat Yuksel
Discover Artificial Intelligence [Q1] [🔗](#)

- [J.7] **EcoStack-Pro: An Adaptive Federated Learning Framework for Interpretable ESG Auditing Across Heterogeneous Industrial Sectors** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Farzan M. Noori, Md Zia Uddin
[Frontiers in Artificial Intelligence \[Q1\]](#) [↗](#)
- [J.8] **Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble with Federated Dynamic Aggregation** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*
[Applied Computational Intelligence and Soft Computing \[Q1\]](#) [↗](#)
- [J.9] **Beyond Short-Term Noise: Robust Long-Horizon Bankruptcy Prediction Using Gated Dual-Branch Transformers** [Paper]
Rezwanul Huque Khan, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Faisal Md. Abdur Rahman, Abdul Kadar Muhammad Masum
[Applied Computational Intelligence and Soft Computing \[Q1\]](#) [↗](#)
- [C.1] **FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification** [Paper]
Md. Abdur Rahman, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay
[International Conference on Data Science, AI and Applications \(ICDSAIA 2025\)](#) [↗](#)
- [C.2] **X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Mohammad Mahmudur Rahman, Iffat Sania Hossain, Abu Kowshir Bitto, Prof. Dr. Abdul Kadar Muhammad Masum
[28th International Conference on Computer and Information Technology \(ICCIT 2025\)](#) [↗](#)
- [C.3] **DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution** [Paper]
Progga Parmita Roy, Sabbir Sarder Munna, Md. Sazed Ul Karim, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Md. Joseb Siddique Siam
[7th IEEE International Conference on Sustainable Technologies for Industry 5.0 \(STI 2025\)](#) [↗](#)
- [W.1] **Do LLMs Distinguish Between Halal and Haram? Benchmarking Islamic Cultural Alignment in General vs. Arabic-Centric SLMs** [Paper]
Md Abdur Rahman, Md Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- [W.2] **Tokenizer Transplantation: Mitigating Autoregressive Collapse in Edge-Efficient Bengali ASR** [Paper]
Sanjid Hasan, *Md Abdur Rahman*
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- [W.3] **SFL-MTKD++: Split ViT Federated Learning with Adaptive Multi-Teacher Distillation for Interpretable Malaria Diagnosis** [Paper]
Md Tofael Ahmed Bhuiyan, *Md Abdur Rahman*, Abdul Kadar Muhammad Masum
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- [W.4] **AlzFed-XAI: High-Fidelity Interpretable Alzheimer’s Diagnosis with Privacy-Preserving Federated Learning** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[5th Muslims in ML Workshop, NeurIPS 2025](#) [↗](#)
- [W.5] **ChakmaBridge: A Five-Way Parallel Corpus for Navigating the Script Divide in an Endangered Language** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[2nd Bangla Language Processing \(BLP\) Workshop, AACL-IJCNLP 2025](#) [↗](#)
- [W.6] **BhasaBodh: Bridging Bangla Dialects and Romanized Forms through Machine Translation** [Paper]
Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Abdul Kadar Muhammad Masum
[2nd Bangla Language Processing \(BLP\) Workshop, AACL-IJCNLP 2025](#) [↗](#)
- [W.7] **Detecting Manipulation in Ukrainian Telegram: A Transformer-Based Approach to Technique Classification and Span Identification** [Paper]
Md. Abdur Rahman, Md Ashiqur Rahman
[4th Ukrainian Natural Language Processing Workshop \(UNLP\), ACL 2025](#) [↗](#)

Under Review Publications

1. HA-DynAU-Net: A Hybrid Attention Dynamic Aware U-Net for Precise Skin Lesion Segmentation
[1st Author]
Submitted to *IEEE Open Journal of the Computer Society*

2. BEACON: A Novel Cross-Modal Architecture for High-Precision and Explainable Brain Tumor Classification Using Medical-Specific Attention
[1st Author]
Submitted to *PLOS ONE*

3. HybridTransUNet: Fusing Vision Transformers with Attentive U-Net for Accurate and Interpretable Brain Tumor Segmentation
[1st Author]
Submitted to *Engineering Reports*

4. Explainable Artificial Intelligence and Deep Learning for Robust Multi-Class Brain Tumor Classification in Medical Imaging
[2nd Author]
Submitted to *Scientific Reports*

Experience with Research Grant

- Assisted in proposal preparation for the following research grants:
- o Closing the Employability Gap: A Machine Learning–Driven Framework to Predict and Improve Student Placement Outcomes at University
Co-PI, IRT Research Grant, Southeast University
Funded amount: \$4,000

o Bridging the Gap: A Predictive Analytics Framework for Enhancing Engineering Student Employability and Placement Outcomes in Bangladesh
Team Member, Bangladesh Accreditation Council (BAC) Research Grant
Funded amount: \$3,000

Honors & Awards

- o 3rd Position (Poster Presentation), Student Research Day 2025

o Champion, Fragments of Feeling: A Contextual Emotion Challenge (2025)

o 1st Runners-up, CalorieQuest: Predicting Energy Burn with Data (2025)

o 2nd Runners-up, DU AI Challenge (2025)

o 2nd Runners-up, REACT 2026 Datathon, hosted by IEEE SEU Student Branch

o Finalist (Top 10), CUET ETE Datathon 2025
- [Certificate]

[GitHub]

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Academic Service

- o Program Committee, 21st BEA Workshop co-located with ACL 2026

o Reviewer, 7th Muslims in ML Workshop at NeurIPS 2026

o Reviewer, Frontiers in Artificial Intelligence

o Reviewer, Frontiers in Medicine

o Reviewer, Green Technologies and Sustainability

Extracurricular Activities

- Research Executive (ExCom), IEEE Computer Society Bangladesh Chapter Graduate Student Member Forum (ExCom)

Sub-Executive (Research & Development), IEEE Computer Society Southeast University Student Branch Chapter
- 2026

2025

Skills

- Languages: C, C++, Java, Python
- Frameworks: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV, Django, FastAPI, Spring Boot
- Technologies: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Git/GitHub, Cloud Computing, LaTeX

Professional Certification

- o AWS Certified Cloud Practitioner, Amazon Web Services

o GitHub Foundations Certification, GitHub
- Dec 2025 – Dec 2028

Nov 2024 – Nov 2027